Language Agent Tree Search

One frozen model as a planning agent

Weekend 6 · Frontier Agentic AI

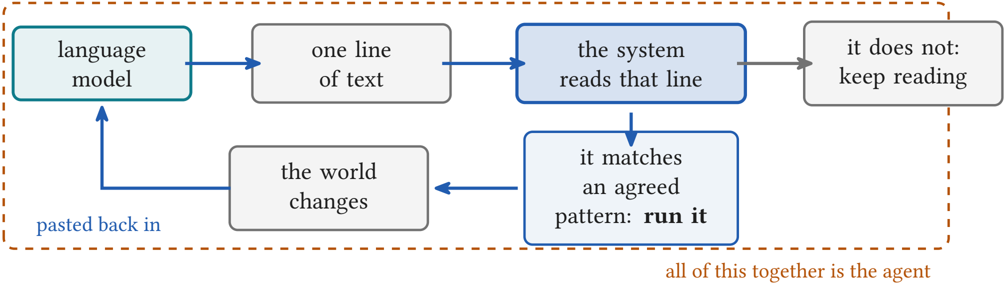
Carlos Cotrini

Weekend 6 · Frontier Agentic AI

Where LATS sits

agents	**LATS**	all of them, inside a search	**we are here**
	Monte Carlo Tree Search	look ahead before moving	this morning
	ReAct	think, act, observe	
prompting	Tree-of-Thought	branch, score, keep the best	
	Chain-of-Thought	write the reasoning first	
the model	Reasoning model	trained by RL to reason before answering	
	Large language model	a transformer compresses any prefix	
	Language model	predict the next word	

How a model gets to act



- The model still only ever emits text. It touches nothing
- The system agrees in advance which text counts as a command

Monte Carlo Tree Search, before language models

1997

Chess falls

Deep Blue beats Kasparov, with search and an evaluation a human wrote

Monte Carlo Tree Search, before language models

1997

Chess falls

Deep Blue beats Kasparov, with search and an evaluation a human wrote

2006

MCTS

no one could write that evaluation for Go, so sample it instead

Monte Carlo Tree Search, before language models

1997

Chess falls

Deep Blue beats Kasparov, with search and an evaluation a human wrote

2006

MCTS

no one could write that evaluation for Go, so sample it instead

2016

Go falls

AlphaGo beats Lee Sedol 4 to 1: the same search, two networks inside it

Monte Carlo Tree Search, before language models

1997

Chess falls

Deep Blue beats Kasparov, with search and an evaluation a human wrote

2006

MCTS

no one could write that evaluation for Go, so sample it instead

2016

Go falls

AlphaGo beats Lee Sedol 4 to 1: the same search, two networks inside it

2018

AlphaZero

the same recipe, learned from self play, no human games

Monte Carlo Tree Search, before language models

1997	2006	2016	2018	today
Chess falls	**MCTS**	**Go falls**	**AlphaZero**	**LATS**
Deep Blue beats Kasparov, with search and an evaluation a human wrote	no one could write that evaluation for Go, so sample it instead	AlphaGo beats Lee Sedol 4 to 1: the same search, two networks inside it	the same recipe, learned from self play, no human games	the same search again, a language model inside it

Monte Carlo Tree Search, before language models

1997

Chess falls

Deep Blue beats Kasparov, with search and an evaluation a human wrote

2006

MCTS

no one could write that evaluation for Go, so sample it instead

2016

Go falls

AlphaGo beats Lee Sedol 4 to 1: the same search, two networks inside it

2018

AlphaZero

the same recipe, learned from self play, no human games

today

LATS

the same search again, a language model inside it

The search itself barely changed across twenty years. What changed each time is what supplies the two things it needs: something to propose moves, and something to judge a position. LATS supplies both from one frozen model.

Coulom 2006; Kocsis and Szepesvari 2006; Silver et al., Nature 2016; Silver et al., Science 2018.

Section 1

The LATS method

1

UCB

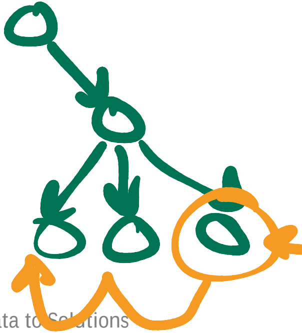
Exploration $\longrightarrow$ inverse to the # of times I have tried it

Exploitation $\longrightarrow$ V(machine)

$$UCB(\text{machine}) = V + \sqrt{\frac{1}{\#\text{times}}}$$

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$



UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

The critic

what the model thinks this branch is worth

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

The critic
what the model thinks
this branch is worth

$+$

The exploration bonus
how little we have tried it

UCT: the critic, plus an exploration bonus

$$\mathrm{UCT}(s) = \underbrace{V(s)}_{\text{the critic}} + \underbrace{w\sqrt{\frac{\ln N(p)}{N(s)}}}_{\text{the exploration bonus}}$$

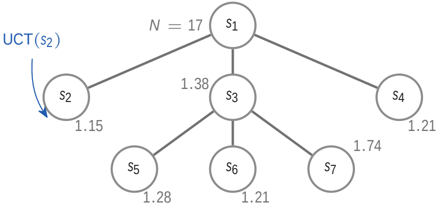
The critic
what the model thinks
this branch is worth

+

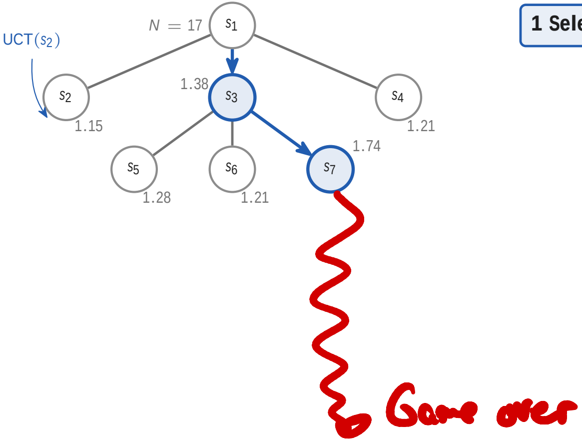
The exploration bonus
how little we have tried it

This morning's words for the same two terms: how good it looked, plus how little we know.

The same loop as this morning, with four edits

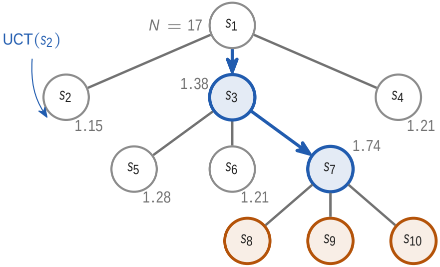


The same loop as this morning, with four edits



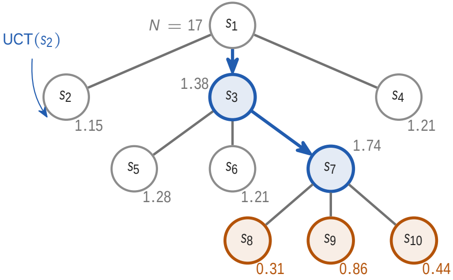
1 Selection by UCT

The same loop as this morning, with four edits



1 Selection	by UCT
2 Expansion	the *actor* proposes

The same loop as this morning, with four edits

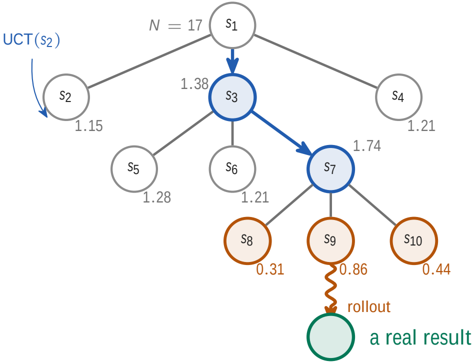


1 Selection by UCT

2 Expansion the *actor* proposes

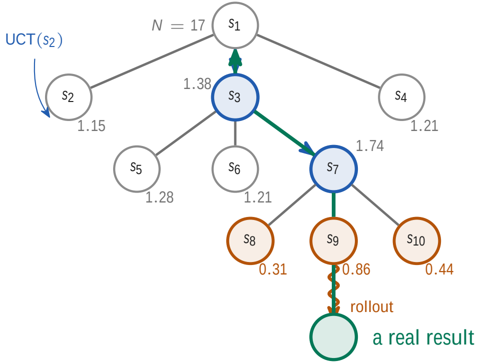
3 Evaluation the *critic* scores

The same loop as this morning, with four edits



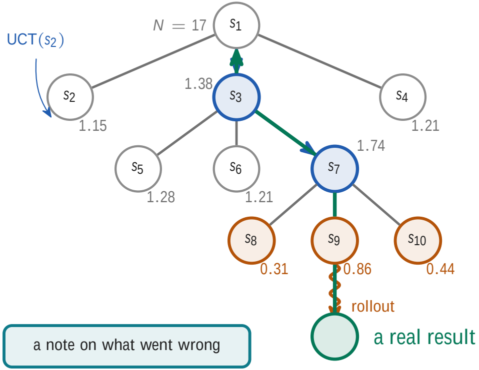
1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end

The same loop as this morning, with four edits



1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end
5 Backpropagation	the *real* result, up

The same loop as this morning, with four edits



1 Selection by UCT

2 Expansion the *actor* proposes

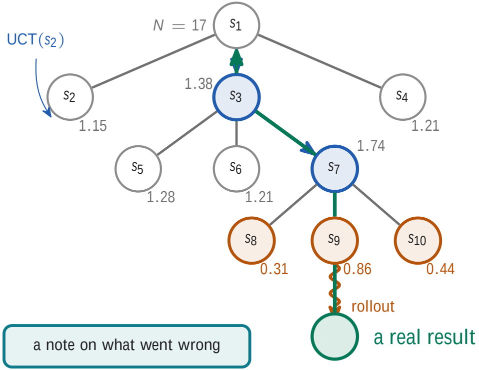
3 Evaluation the *critic* scores

4 Simulation the *best* one, to the end

5 Backpropagation the *real* result, up

6 Reflection the *feedback* model writes it

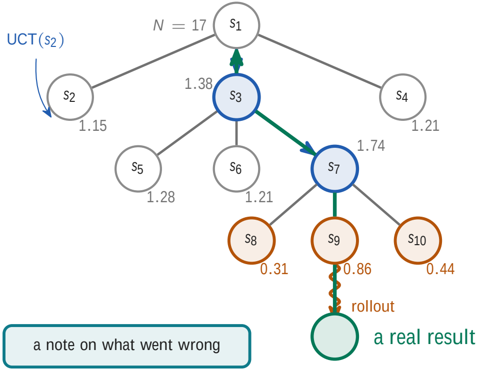
The same loop as this morning, with four edits



1 Selection	by UCT
2 Expansion	the *actor* proposes
3 Evaluation	the *critic* scores
4 Simulation	the *best* one, to the end
5 Backpropagation	the *real* result, up
6 Reflection	the *feedback* model writes it

grey: UCT, what selection compares.
amber: *V* from the critic, and all that is new.

The same loop as this morning, with four edits

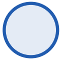


- **1 Selection** — by UCT
- **2 Expansion** — the *actor* proposes
- **3 Evaluation** — the *critic* scores
- **4 Simulation** — the *best* one, to the end
- **5 Backpropagation** — the *real* result, up
- **6 Re□ection** — the *feedback* model writes it

into the next prompt

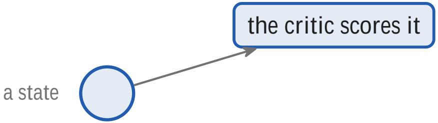
grey: UCT, what selection compares.
amber: V from the critic, and all that is new.

What the critic actually returns

a state 

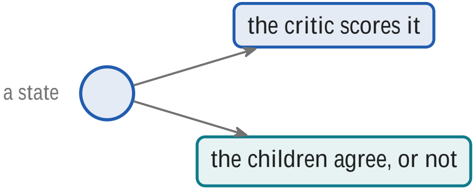
$$V(s) = \qquad +$$

What the critic actually returns



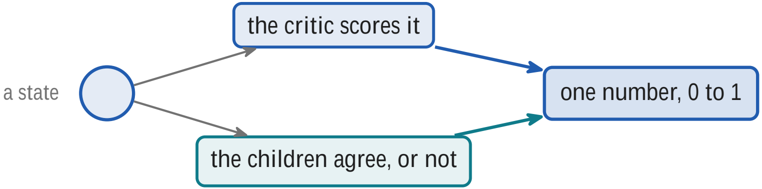
$$V(s) = \qquad +$$

What the critic actually returns



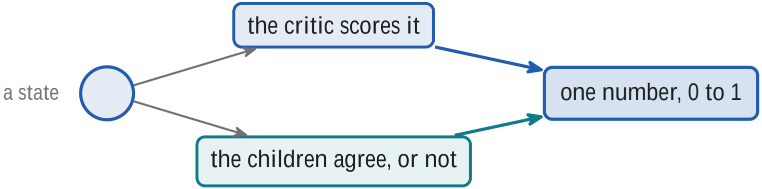
$$V(s) \;=\; \qquad\qquad + $$

What the critic actually returns



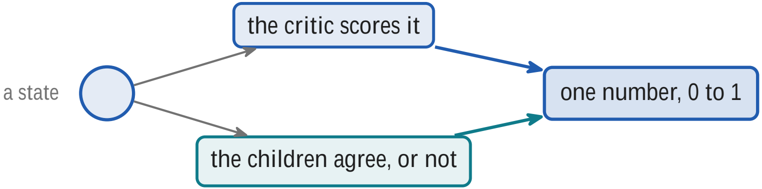
$$V(s) \; = \qquad\qquad + $$

What the critic actually returns



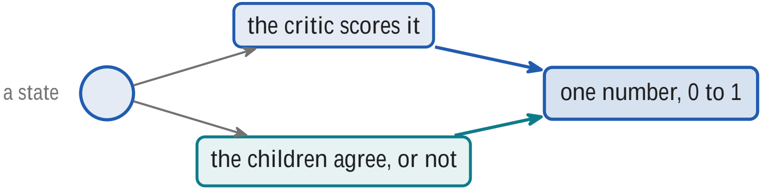
$$V(s) = \lambda \mathrm{LM}(s) +$$

What the critic actually returns



$$V(s) = \lambda\, \mathrm{LM}(s) + (1-\lambda)\, \mathrm{SC}(s)$$

What the critic actually returns



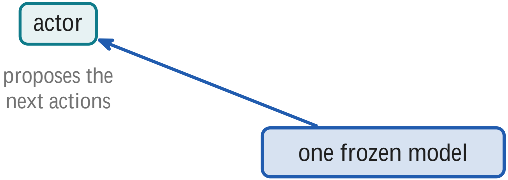
$$V(s) = \lambda\,\mathrm{LM}(s) + (1-\lambda)\,\mathrm{SC}(s)$$

$\lambda = 0.5$ on question answering, 0.8 on programming and web tasks.

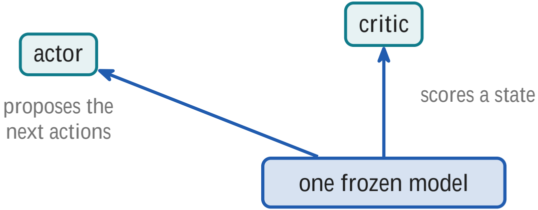
Three jobs, one set of weights

one frozen model

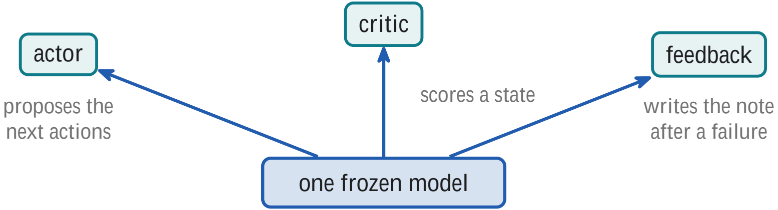
Three jobs, one set of weights



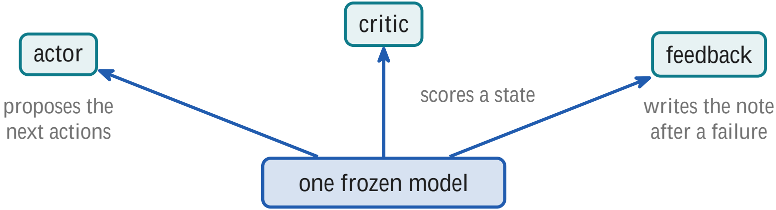
Three jobs, one set of weights



Three jobs, one set of weights



Three jobs, one set of weights



same weights every time. Only the prompt changes.